

KUSHAGRA SINGH TIWARI

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PROFESSIONAL SUMMARY

Data Engineer / Python Developer with **2+ years** of experience building scalable **ETL/ELT pipelines**, automating data workflows, and developing AI-powered applications. Experienced in **Python, SQL, Databricks, Oracle, PostgreSQL, PySpark**, and Generative AI technologies including **LangChain, RAG**, and **vector databases**. Passionate about combining data engineering with AI/ML to build intelligent, production-ready data solutions.

CORE SKILLS

Languages: Python, Java, SQL, PL/SQL, NoSQL, Shell Scripting

Data Engineering: PySpark, Apache Spark, Delta Lake, Medallion Architecture, ETL/ELT, Data Warehousing, Data Modeling, Data Validation, Workflow Automation, REST APIs, SFTP

AI/GenAI: LangChain, CrewAI, LangGraph, Gemini, RAG, Vector Databases, FAISS, Chroma, Prompt Engineering, OCR, LLMs, MCP

Databases: Oracle, PostgreSQL, MySQL, SQLite, Azure Databricks, Snowflake

Cloud & DevOps: Git, Linux, CI/CD, AWS, GCP, Azure, Terraform

EXPERIENCE

Splash Business Intelligence Pvt. Ltd. | *Associate Software Engineer*

Feb 2024 – Present | Hyderabad

- Engineered high-performance **PL/SQL procedures** and ETL pipelines across **Oracle Cloud, UKG, SAP**, and **PostgreSQL**, automating **REST API-based currency exchange data** integration and improving data processing efficiency by **25%**.
- Designed Python-based file ingestion and **SFTP automation workflows** for enterprise ETL pipelines, reducing manual processing effort by **90%**.
- Developed **PySpark-based validation frameworks** in **Databricks** to verify incremental ETL loads, ensuring data consistency and accuracy across **Bronze, Silver**, and **Gold layers** for multiple enterprise modules.
- Implemented a **centralized telemetry automation pipeline** for **40+ clients**, consolidating ETL execution metrics into a single dataset, loading it into a database, and enabling **automated email alerts and reporting**.
- Contributed to the company's **SplashAI** platform by developing AI-powered document processing and automation solutions using **CrewAI, Gemini API, OCR, and LLMs**, improving intelligent data extraction and workflow efficiency.

PHN Technology Pvt. Ltd. | *Programming Intern*

04/2023 – 06/2023 | Pune

- Developed a **Python-based machine learning model** using **NumPy, Pandas, Scikit-learn, Matplotlib, and OpenPyXL** to predict linear relationships from structured datasets.
- Built and optimized a **data preprocessing and feature engineering pipeline**, improving model accuracy by **~15%**.

PROJECTS

Local AI Agent (RAG-Based Intelligent Q&A System)

Nov 2025 – Dec 2025

- Built a **locally deployed AI agent** using **Retrieval-Augmented Generation (RAG)** for contextual document-based Q&A.
- Integrated **Ollama** to run LLMs locally, enabling **offline and private inference** without external APIs.
- Implemented a **Chroma-based vector search pipeline** for fast semantic document retrieval.
- Orchestrated LLM workflows using **LangChain** and optimized document preprocessing with **Pandas**.
- Tech Stack:** Python, LangChain, Ollama, FAISS, Pandas

Image Forgery Detection & Localization

Aug 2023 – Dec 2023

- Developed a **U-Net-based 2D CNN model** to detect copy-move and splicing image forgeries.
- Performed **image preprocessing, annotation handling, and dataset preparation** to improve model robustness and generalization. Achieved **76% Binary IoU**, demonstrating strong localization performance.
- Tech Stack:** Python, Tensorflow, Keras

EDUCATION

Master of Computer Applications (MCA) | University of Hyderabad, Hyderabad

2022 - 2024

Bachelor of Science | APS University, Rewa, Madhya Pradesh

2018 - 2021